

Kyle Mason

Network & Infrastructure Engineering

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Summary

Professional helicopter pilot, flight instructor, and flight-safety (FOQA) officer — and a U.S. Marine Corps infantry veteran — transitioning into **network and infrastructure engineering**, bringing eight years of fleet-scale telemetry analysis, anomaly detection, root-cause tracing, procedure standardization, and personnel training to infrastructure engineering with the same checklist-driven, zero-defect discipline. CCNA in progress. Builder and operator of NETFRAME, a 7-node compute platform (below), and contributor to Prometheus node_exporter.

Technical Skills

- **Networking:** VLAN segmentation, L2/L3 routing & switching (Juniper EX / JunOS), OPNsense firewalling, DNS/DHCP HA, 10 GbE, network-as-code (containerlab, FRR/OSPF)
- **Infrastructure & Virtualization:** Proxmox VE clustering, RKE2 Kubernetes (Cilium, MetalLB), ZFS, Proxmox Backup Server, SLURM (multi-tenant GPU scheduling)
- **Automation, IaC & Platforms:** Ansible, Bash, Python, Git, GitHub Actions CI/CD; Linux (Debian/Proxmox), Docker/Apptainer, systemd, self-hosted LLM inference (Ollama)
- **Observability & Security:** Prometheus, Grafana, Loki, Wazuh SIEM, firewall/VLAN hardening, Headscale VPN, step-ca (internal PKI/TLS)

Projects

NETFRAME — Research & Education Compute Cluster · *self-directed* · github.com/machismo0311/Home-Lab
7-node Proxmox cluster — ~140 cores, 3 GPUs / 104 GB VRAM, ~95 TB ZFS — *designed, built, and operated end-to-end.*

- Engineered a **7-VLAN segmented network** on a Juniper EX3400 core + OPNsense firewall, with **HA DNS/DHCP failover** and a 10 GbE fabric to compute/storage nodes.
- Built a **multi-tenant GPU environment** (SLURM + NVIDIA MPS + Apptainer) sharing one GPU across a researcher's production workload and ~15 students/semester — hard per-job VRAM caps, container isolation, and priority preemption.
- Stood up **RKE2 Kubernetes** (3-node HA control plane, Cilium, MetalLB, private TLS registry), self-hosted **Headscale VPN**, **Wazuh SIEM**, Grafana→Discord alerting, **tested backups** (PBS, 7d+4w, restore-verified), and CI/CD across repos.

Self-testing virtual network (netlab) · [Home-Lab/netlab](https://github.com/machismo0311/netlab)

Built a **containerlab** FRR/OSPF topology that GitHub Actions **deploys and validates on every push** — asserting real end-to-end reachability and control-plane state. Diagram-as-code regenerates the network diagram from a source-of-truth inventory (CI fails on drift).

netframe-monitor · github.com/machismo0311/netframe-monitor

Read-only cluster health monitor — SSH diagnostics + local-LLM interpretation + a web report on a systemd timer.

Open Source Contributions

Prometheus node_exporter · PR #3784 · *open, under review*

Fixed a Linux collector bug where duplicate kernel slab-cache names produced duplicate Prometheus label sets and failed the whole metrics scrape; added an index label per exporter convention, four regression tests, and updated end-to-end fixtures.

Professional Experience

Metro Aviation, Inc.

Nationwide, U.S. · 2018 – Feb 2026

Check Airman · EC-135/145 Instructor Pilot · Senior FOQA Officer

- **Flight-data telemetry analysis (FOQA):** owned analysis and security of fleet-scale flight-data telemetry; detected anomalies and exceedances, traced findings to root cause, and drove corrective actions through a non-punitive (blameless) safety program improving training and procedures.
- **Training & standardization:** conducted and evaluated pilot training and checkrides under a Part 135 program; ensured FAA compliance and standardized procedures; developed the training program; onboarded and mentored new-hire pilots and instructors.

United States Marine Corps

Camp Lejeune, NC · 2003 – 2007

Infantry Squad Leader

- Led a 12-Marine squad — training, discipline, and safety in austere environments worldwide, including **two combat deployments to Iraq**.
- Operated effectively under pressure in diverse, resource-constrained conditions, with full accountability for the squad and mission.

Education & Certifications

M.S., Information Systems — Cleveland State University

planned

B.S., Aerospace Science — Embry-Riddle Aeronautical University

2019

CCNA (Cisco) — *in progress* · FAA Commercial Helicopter Pilot — Instrument Rated; CFI/CFII